Ikeda–Sine–Chebyshev Map

Derivation and Numerical Dynamical Analysis

a = 0.9, c = 0.4, d = 6.0; ω = 10 for phase-space and Shannon analysis

1. Derivation of the Map

The proposed Ikeda–Sine–Chebyshev (ISC) map is formed by nesting three nonlinear operations: the two-dimensional Ikeda transformation, a sinusoidal transformation, and the second-order Chebyshev polynomial. The map remains strictly two-dimensional: x_n and y_n are the only evolving variables.

1.1 Ikeda stage

For state (x_n, y_n), first define the nonlinear Ikeda angle:

$$t_n = c - d/(1 + x_n^2 + y_n^2)$$

$$X_n = 1 + a[x_n \cos(t_n) - y_n \sin(t_n)]$$

$$Y_n = a[x_n \sin(t_n) + y_n \cos(t_n)]$$

1.2 Sine stage

The Ikeda intermediate coordinates are then passed through a sinusoidal layer:

$$S_x = \sin(\omega X_n)$$

$$S_y = \sin(\omega Y_n)$$

1.3 Chebyshev stage

The second-order Chebyshev polynomial is

$$T_2(z) = 2z^2 - 1$$

Applying T_2 to the sine outputs gives

$$x_{n+1} = 2\sin^2(\omega X_n) - 1$$

$$y_{n+1} = 2\sin^2(\omega Y_n) - 1$$

Overall structure: Ikeda → Sine → Chebyshev

Substituting X_n and Y_n, the complete map is

$$x_{n+1} = 2\sin^2\{\omega[1 + a(x_n \cos t_n - y_n \sin t_n)]\} - 1$$

$$y_{n+1} = 2\sin^2\{\omega a(x_n \sin t_n + y_n \cos t_n)\} - 1$$

$$t_n = c - d/(1 + x_n^2 + y_n^2)$$

Since $0 \le \sin^2(\cdot) \le 1$, the final transformation gives $-1 \le x_{n+1}, y_{n+1} \le 1$. Therefore the trajectory is confined to $[-1,1]^2$ after the first iteration. No third state variable or external/random sequence is introduced.

2. Jacobian Used for Lyapunov Exponents

Let

$$u_n = x_n \cos(t_n) - y_n \sin(t_n)$$

$$v_n = x_n \sin(t_n) + y_n \cos(t_n)$$

$$t_x = 2d\, x_n/(1 + x_n^2 + y_n^2)^2$$

$$t_y = 2d\, y_n/(1 + x_n^2 + y_n^2)^2$$

The Ikeda-stage Jacobian is

$$J_I = a\,[[\cos(t_n) - v_n\, t_x,\ -\sin(t_n) - v_n\, t_y],$$

$$[\sin(t_n) + u_n\, t_x,\ \cos(t_n) + u_n\, t_y]]$$

For $f(X) = 2\sin^2(\omega X) - 1$:

$$df/dX = 4\omega \sin(\omega X)\cos(\omega X)$$

$$J = \mathrm{diag}(4\omega \sin(\omega X_n)\cos(\omega X_n),\ 4\omega \sin(\omega Y_n)\cos(\omega Y_n))\, J_I$$

3. Numerical Results

The four figures below are the supplied numerical results for the ISC map. The phase portrait and Shannon entropy use $\omega = 10$; the parameter scans cover $\omega = 0$ to 15.

3.1 Phase-space trajectory

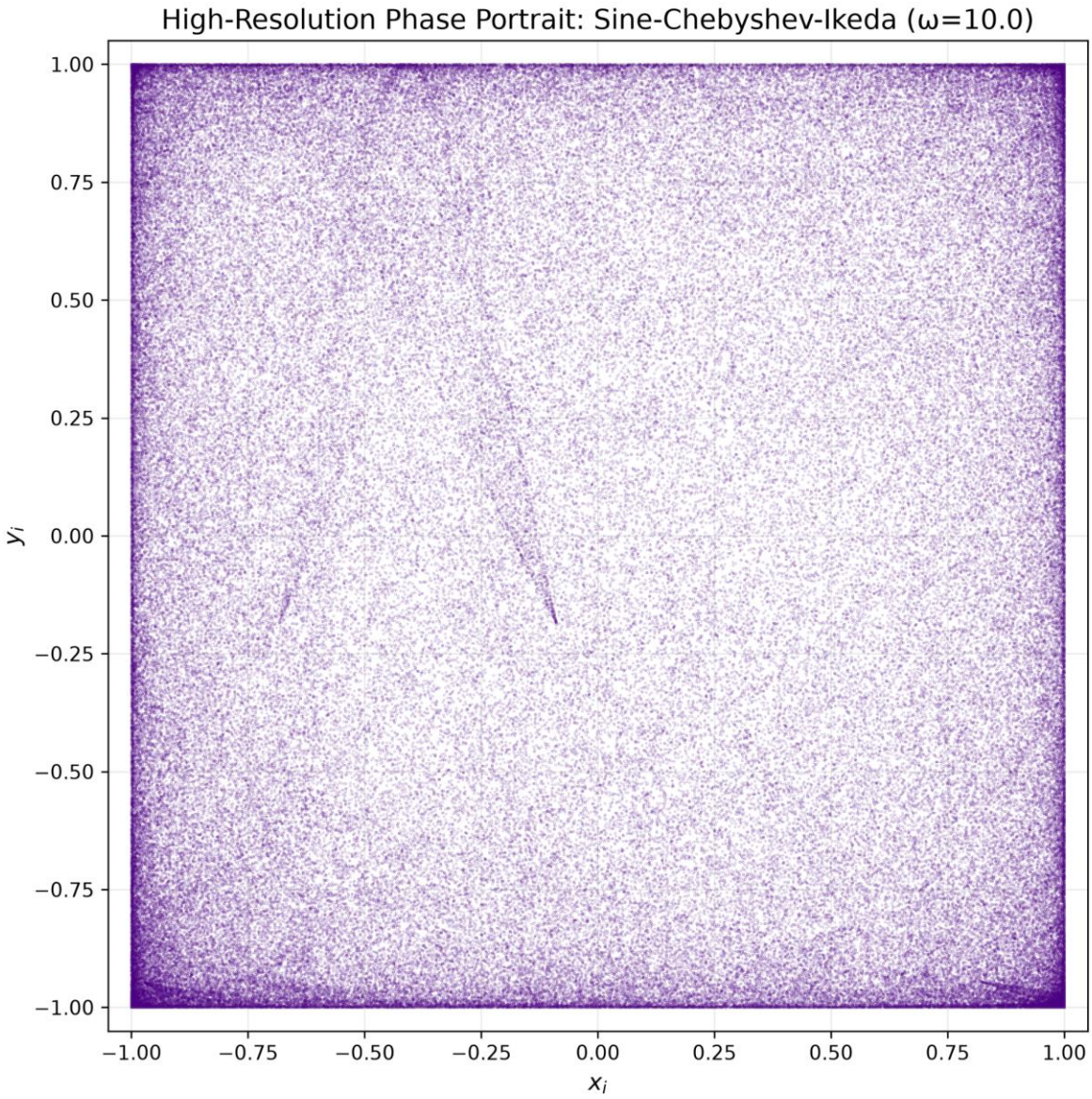
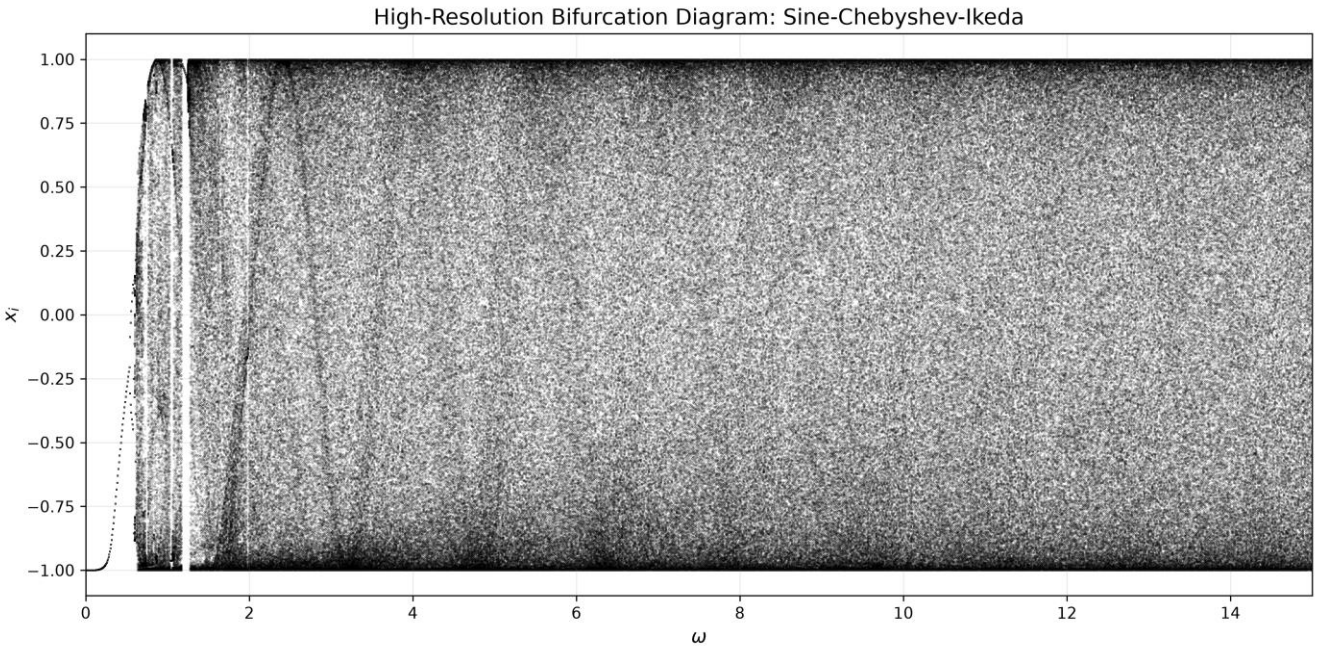


Figure 1. Phase-space trajectory at ω = 10.

At ω = 10, the orbit occupies a large portion of $[-1,1]^2$. The dense distribution indicates broad phase-space occupancy, while nonuniform density reflects the nonlinear folding of the nested map.

3.2 Bifurcation diagram

The bifurcation diagram shows a comparatively regular low-ω regime followed by broad dense parameter regions. The diagram is interpreted together with the Lyapunov spectrum rather than as a standalone proof of chaos.



3.3 Lyapunov exponents

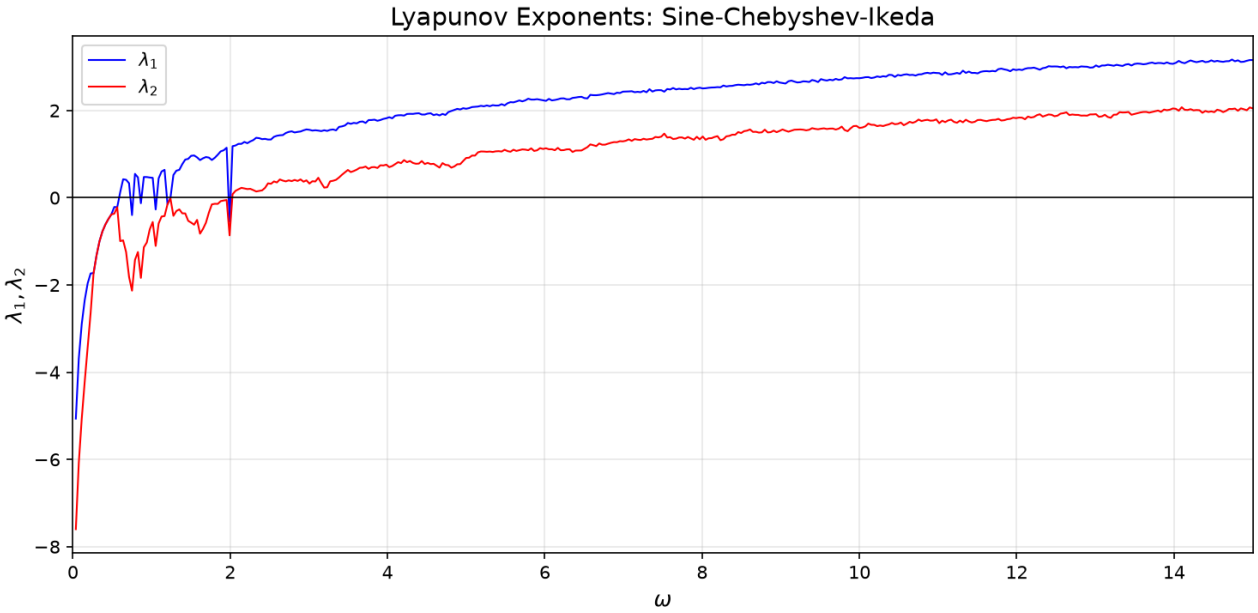


Figure 3. Lyapunov exponents λ_1 and λ_2.

Both Lyapunov exponents become positive over a broad portion of the scanned parameter range. For this two-dimensional map, simultaneous positivity indicates expansion in both tangent-space directions.

3.4 Kolmogorov–Sinai entropy

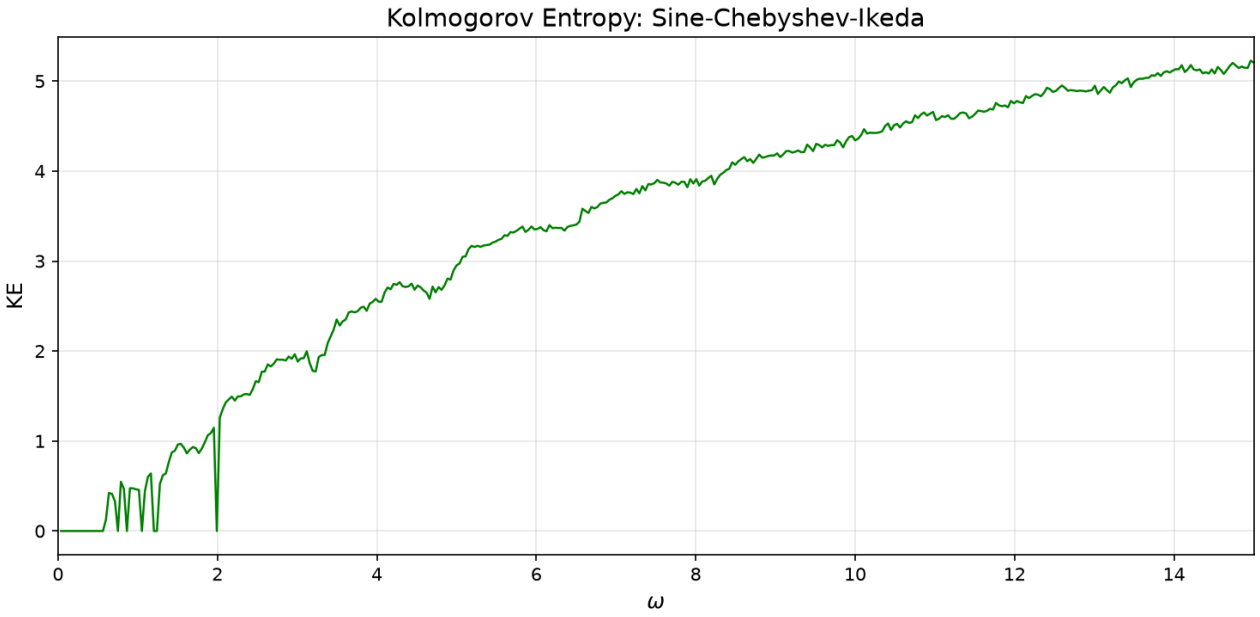


Figure 4. Kolmogorov–Sinai entropy estimate.

The plotted KS-entropy estimate is $h_{KS} = \Sigma \max(\lambda_i, 0)$. Its increase with ω follows the increase in positive Lyapunov exponents.

4. Shannon Entropy Result

A binned Shannon entropy test was performed on the 200,000-point phase-space trajectory at ω = 10. The state space was partitioned into 256 × 256 bins over $[-1,1]^2$. For probabilities p_i,

$$\mathbf{H = -\Sigma\, p_i \log_2(p_i)}$$

The maximum entropy for 256^2 equally probable bins is

$$\mathbf{H_{max} = \log_2(256^2) = 16 \text{ bits}}$$

Metric	Result
H(x)	7.693507 bits
H(y)	7.664964 bits
H(x,y)	15.083315 bits
H_max(x,y)	16.000000 bits
Normalized H(x,y)	0.942707

Normalized H(x,y) = 15.083315 / 16 = 0.942707 ≈ 94.27%

The normalized two-dimensional entropy is therefore approximately 94.27% of the maximum under the selected discretization. This is a high phase-space distribution measure. Because histogram entropy depends on bin resolution, sample count, transient removal, and state-space range, identical settings must be used for any comparison with other maps.

5. Summary

- The ISC construction is strictly two-dimensional and autonomous.
- The Ikeda stage provides state-dependent nonlinear rotation and stretching.
- The Sine stage introduces periodic nonlinear folding.
- The T_2 stage adds the nonlinear transformation $2z^2-1$.
- The final state is bounded in $[-1,1]^2$.
- At $\omega = 10$, the phase portrait shows broad phase-space occupancy.
- The Lyapunov spectrum shows a broad region with two positive exponents.
- The KS entropy increases across the stronger chaotic regime.
- The measured normalized 2-D Shannon entropy at $\omega = 10$ is 0.942707 (94.27%).

6. Methodological Note

These results correspond specifically to the Ikeda → Sine → Chebyshev ordering implemented in the provided code. The Chebyshev operation is the polynomial $T_2(z)=2z^2-1$ applied to the sinusoidal outputs. Before publication, the Lyapunov calculation should be independently checked using the Jacobian-determinant consistency relation, and the Shannon entropy should be tested for stability with respect to sample size and bin resolution.